COLLEGE OF SCIENCE AND TECHNOLOGY

Rinchending, Phuentsholing

Royal University of Bhutan

Programming Assignment-1

Module: SWE201 – Cross Platform Development
Title: Multi-Screen "Campus Companion" App (React Native)

Deadline: 10th April 2026
Assignment Type: Individual programming assignment
Weight: As per module Programming Assignment component (10%)

1. Objective

Design and implement a small but complete React Native mobile application that demonstrates your understanding of:

- Basic React Native architecture, project structure, and core components (View, Text, Image, ScrollView, FlatList).
- Styling with StyleSheet, Flexbox layout, and simple platform-specific or dynamic styles.
- Navigation using React Navigation (stack + at least one other navigator).
- Responsive design for different screen sizes and orientations.

2. Scenario

You are asked to build a **"Campus Companion"** app to help new students at your college quickly access key information (e.g., important contacts, campus map links, class schedule overview, announcements). The app should consist of multiple screens, intuitive navigation, and a clean,

responsive UI suitable for both small and large Android devices. You may later extend this app in future assignments, so design it with basic modularity in mind.

3. Functional Requirements

Your app must satisfy at least the following:

1. **Minimum Screens (4+):**
   - Home screen with app title and navigation entry points.
   - "Contacts" screen listing important contacts (e.g., IT helpdesk, student services) using FlatList.
   - "Schedule/Timetable" screen showing a simple weekly class overview.
   - At least one additional screen of your choice (e.g., "Campus Map Links", "Clubs & Events", "Notice Board").
2. **Navigation:**
   - Use **React Navigation** with a **Stack Navigator** and at least one additional navigator type (Bottom Tabs or Drawer).
   - Pass parameters between screens at least once (e.g., tap a contact to view details on a separate screen).
3. **Styling & Layout:**
   - Use **StyleSheet** and **Flexbox** for layout (no inline styles).
   - Implement at least one simple **dynamic style** (e.g., highlight selected items, light/dark theme toggle, or platform-specific style using `Platform.select`).
4. **Responsive Design:**
   - Use **Dimensions API and/or percentage-based/flex layouts** so that UI scales reasonably on different screen sizes.

- Ensure content is scrollable where appropriate (ScrollView/FlatList) and does not overflow on smaller devices.

You may use either JavaScript or TypeScript, but the project must compile and run on an Android/IOS emulator/simulator or device.

4. Technical & Submission Requirements

- Use the Expo.
- Initialize a **new project** for this assignment and maintain a clean directory structure (separate folders for components/screens where appropriate).
- Commit code to the designated **online version control system** (e.g., Git repository) and submit the repository link alongside report through the **VLE** before the deadline.
- Include a brief `README.md` describing:
  - App overview and main features (Only Overview).
  - Steps to install dependencies and run the app (including `npm/yarn` commands).
  - Any known issues or limitations.
- Submit the **Assignment Report in PDF format** on **VLE** with proper College format (Make it standard and of required page length only).
- Attached all **screenshots** with the report.

5. Assessment Criteria (10 marks total)

- **Code Organization & Readability (2 marks):**
  Clear structure (screens/components separated), meaningful naming, no dead/duplicate code.
- **Code Comments & Documentation (1 mark):**
  Concise comments for non-obvious logic, informative `README.md`.
- **Functional Requirements (5 marks):**
  Correct implementation of required screens, navigation, styling, and responsiveness; app runs without crashes.

- **User Interface & User Experience (1 mark):**
  Visually consistent design, legible typography, intuitive navigation.
- **Deployment & Configuration (1 mark):**
  Project builds and runs on a platform specific emulator/simulator/device with the steps provided in README.md.

Late submissions and academic integrity rules apply as per university regulations.

DO not Use AI Tools for this Assignment.